

Long Distance Power Transmission - Medium Problem

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You are designing a system to send power from a 120V standard portable generator down to a rocket launch site. The power from this generator is critical, as it will be supporting all infrastructure near the rocket. Unfortunately though, you notice an issue. The rocket is 1000 feet away, and the system draws 15 amps. Doing a little research about the extension cord you are using, you find that for 12 gauge wire has a resistance of about 1.5 ohms per 1000 feet. This means that you will have $15 \cdot 1.5 = 22.5V$ of voltage drop on half the round trip. You also have the 1000 feet coming back! This means your ground (neutral wire in 120VAC) also gets 22.5V of drop. As such, your equipment at the rocket sees $120 - 2 \cdot 22.5 = 75V$. This will not work.

Design a circuit that goes on the transmitting and receiving ends of this system to resolve this issue. You are not allowed to change the linear resistance of the cable to the rocket. An acceptable voltage drop is 5V off of 120V at 15 amps (after the power has fully passed through your circuit). This is equivalent to having 75 watts being dissipated in the transmission wire.